Name
Date
Course

Transformations and the Library of Relations (Homework)

Write the parent relation and the parent equation of each relation below and use this information to graph the modified equation. Graphing paper will be useful.

① $y-2=(x+1)^2$

② $|x-3|=y+4$

③ $y+1=\sqrt{x-1}$

④ $(x-4)^3=y-3$

⑤ $y+1=x+2$

⑥ $-y=|x|$

⑦ $y=(-x)^3$

⑧ $-y=x^2$

⑨ $y=\sqrt{-x}$

⑩ $y=|-x|$

⑪ $y=-x$

Write the parent relation and the parent equation of each relation below and use this information to graph the modified equation. Then, write the type of dilation involved.

⑫ $y=|2x|$

⑬ $3y=\sqrt{x}$

⑭ $y=(3x)^3$

⑮ $2y=x^2$

⑯ $y=2x$

⑰ $y=\left|\frac{x}{3}\right|$

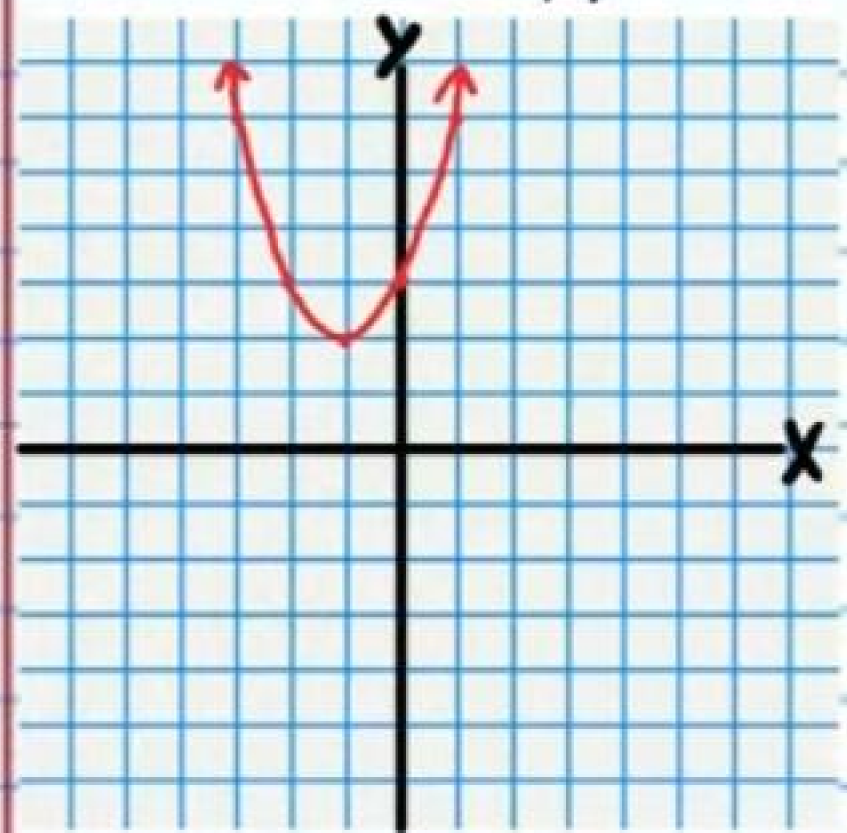
⑱ $\frac{y}{4}=\sqrt{x}$

⑲ $y=\left(\frac{x}{3}\right)^3$

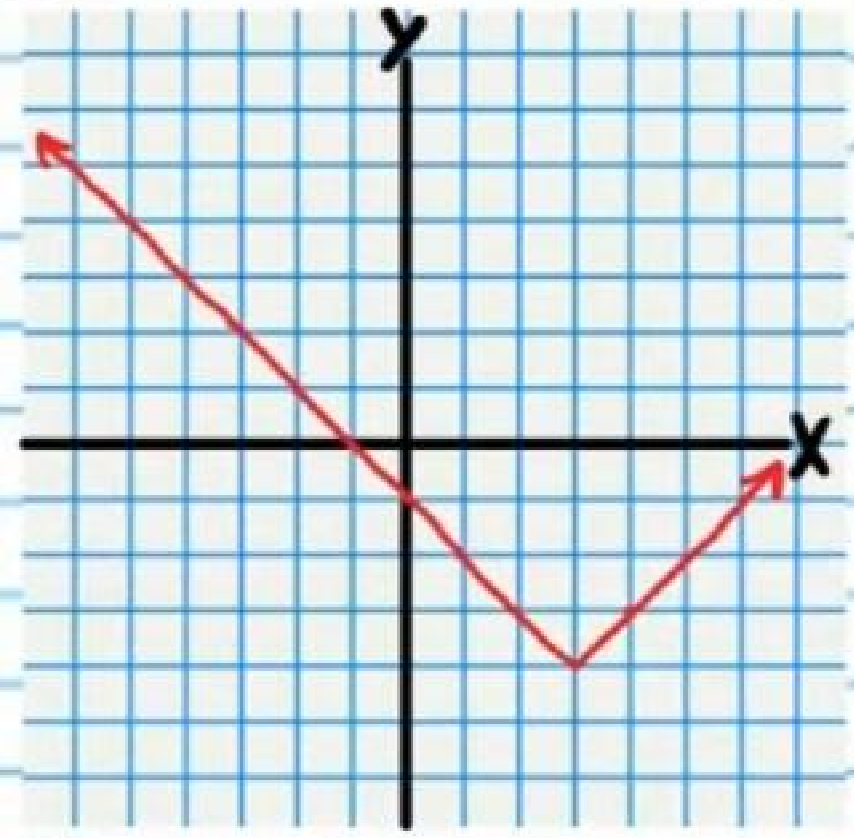
⑳ $\frac{y}{5}=x^2$

Answers

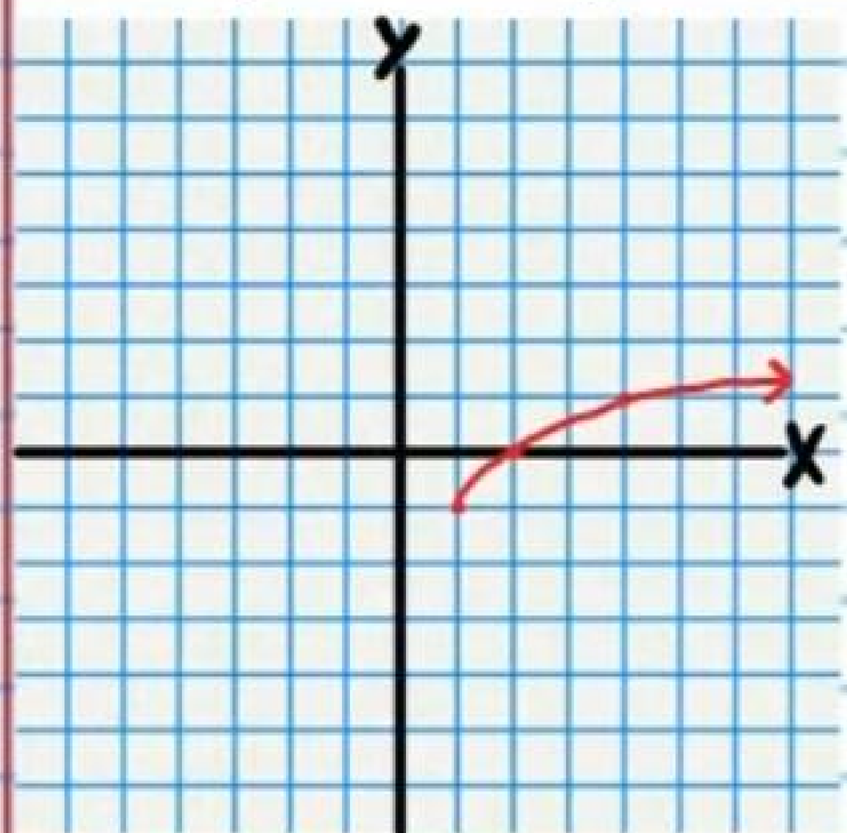
① Quadratic, $y=x^2$



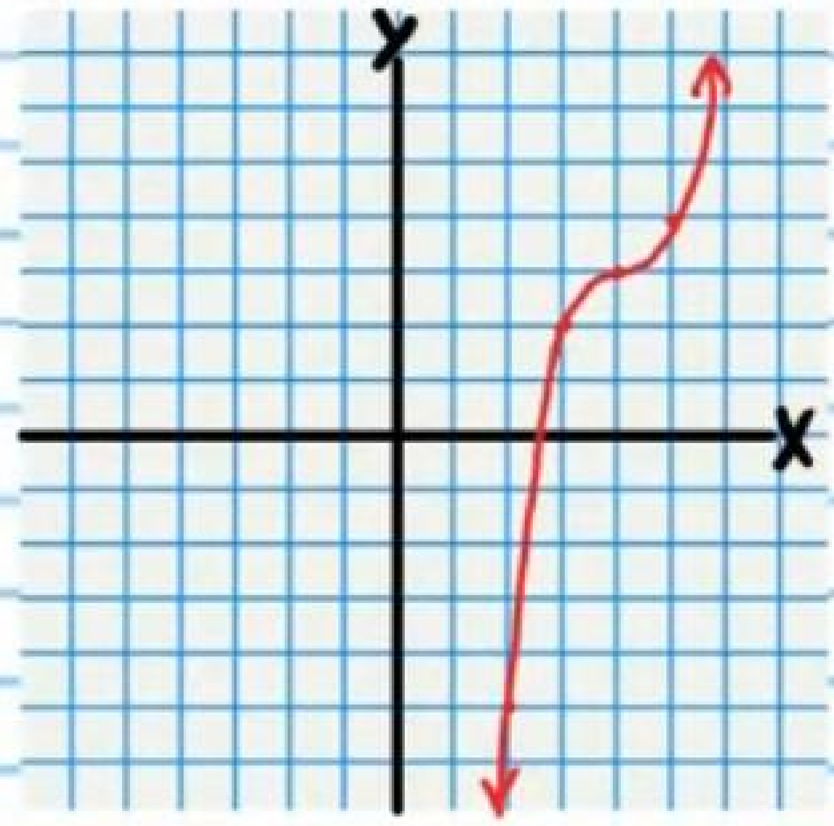
② Absolute Value, $y=|x|$



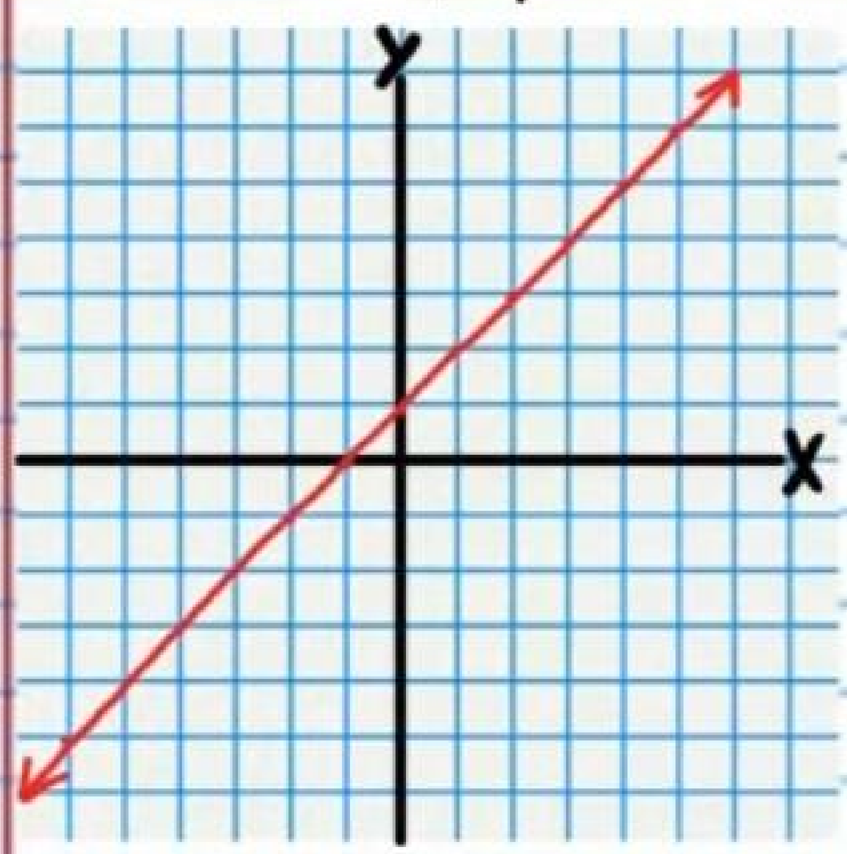
③ Square Root, $y=\sqrt{x}$



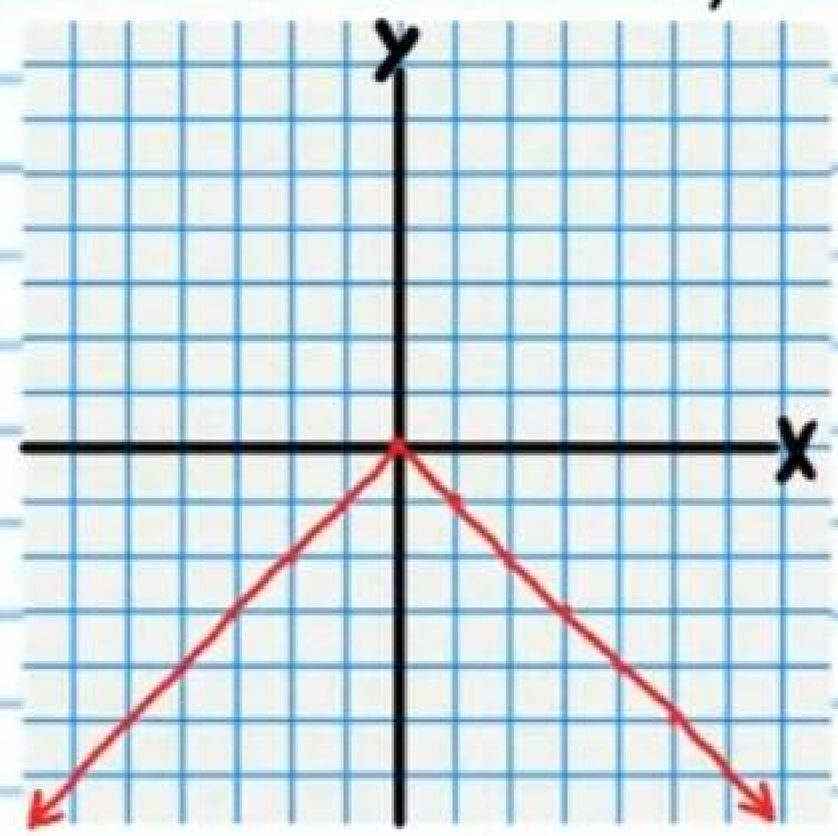
④ Cubic, $y=x^3$



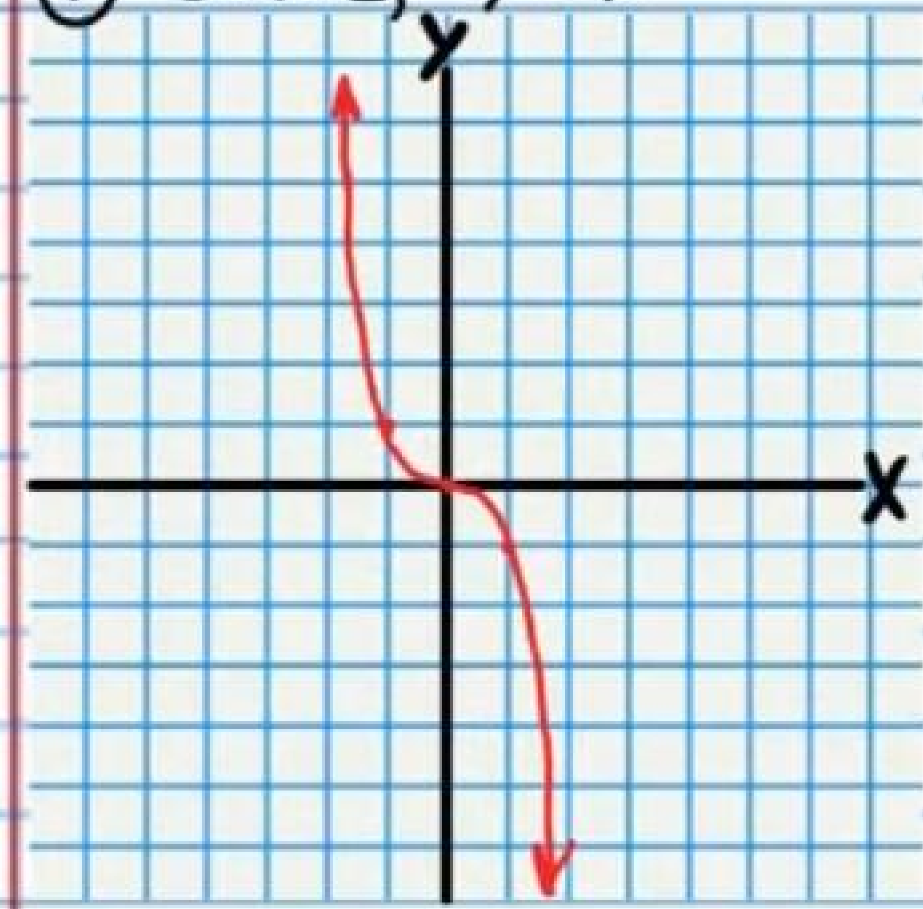
⑤ Linear, $y=x$



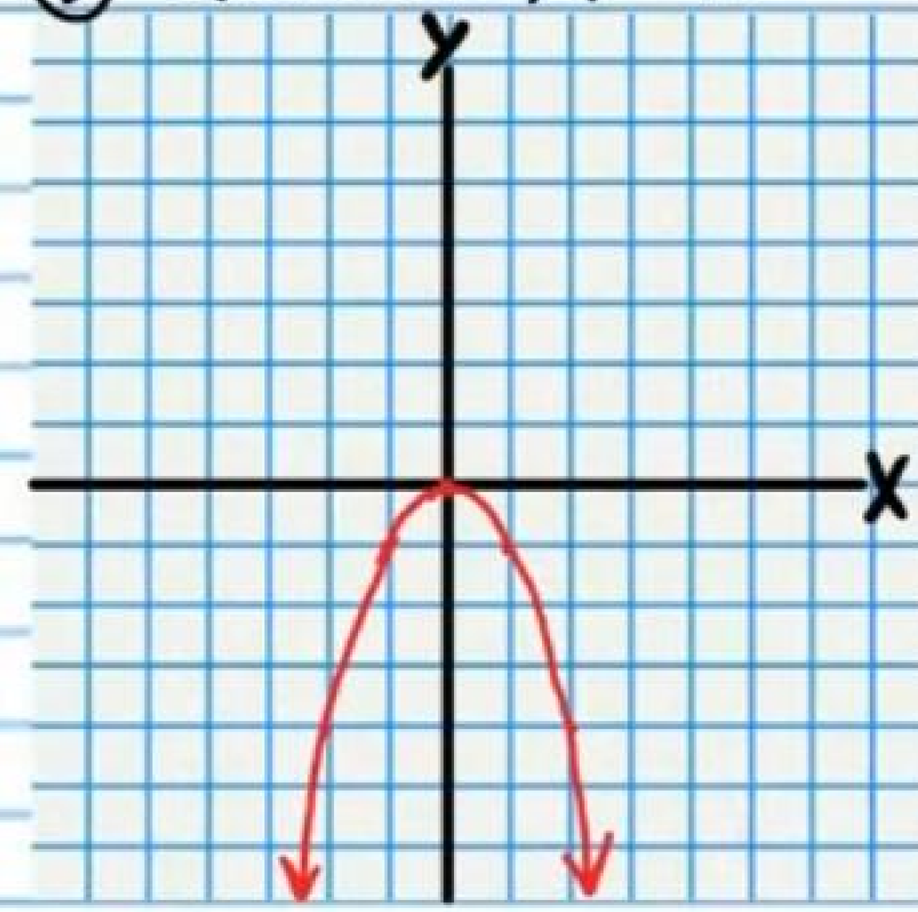
⑥ Absolute Value, $y=|x|$



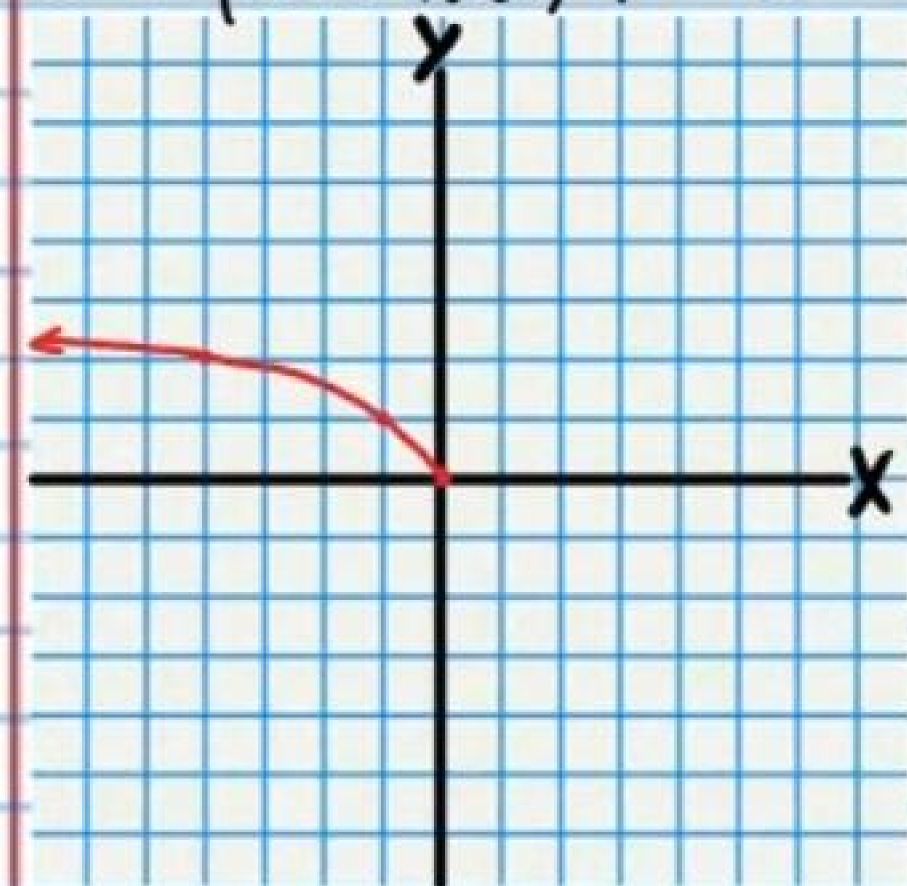
⑦ Cubic, $y=x^3$



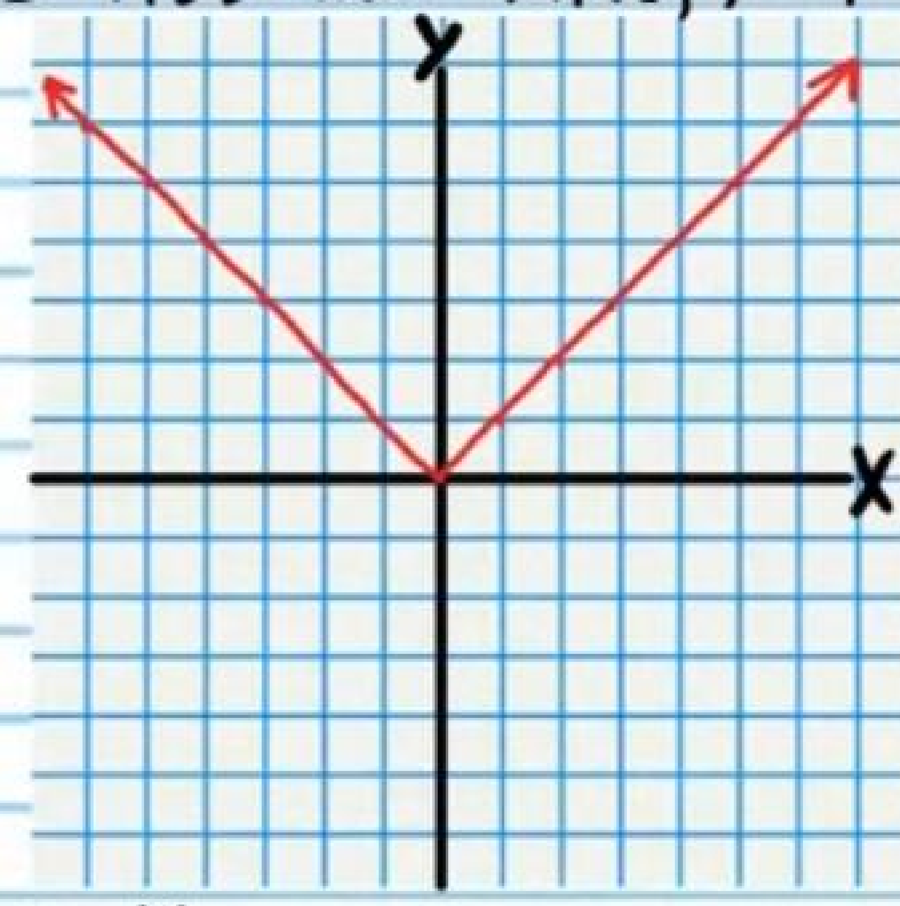
⑧ Quadratic, $y=x^2$



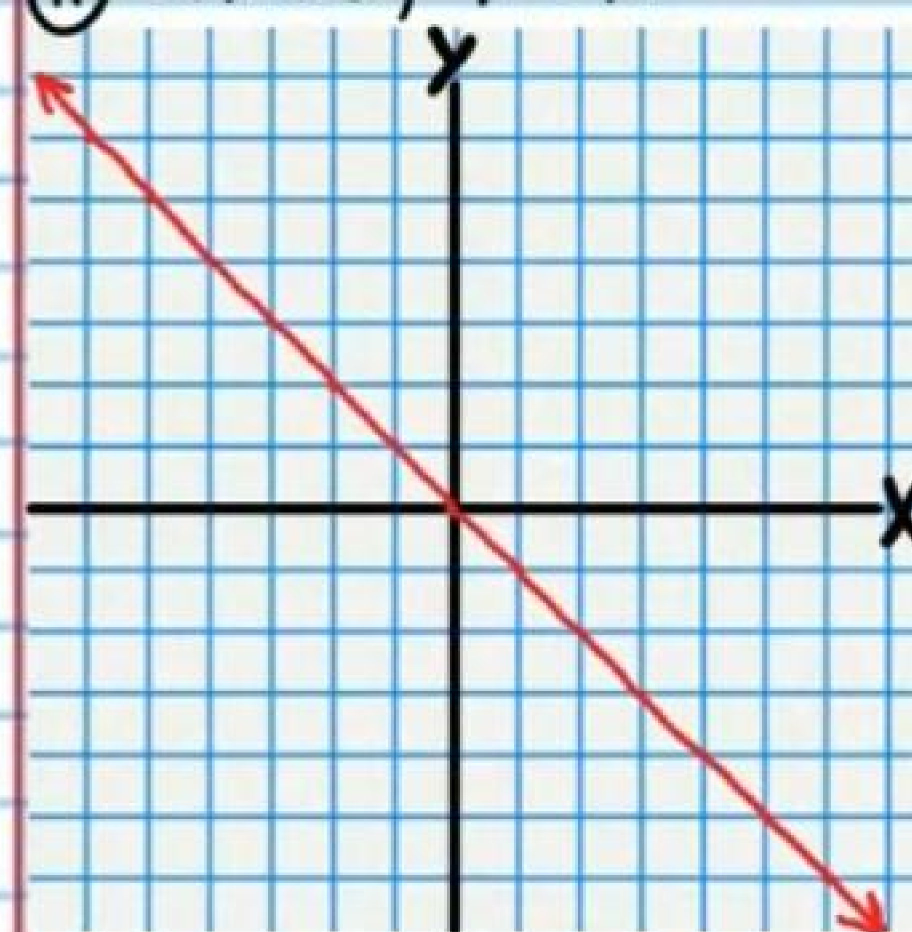
⑨ Square Root, $y=\sqrt{x}$



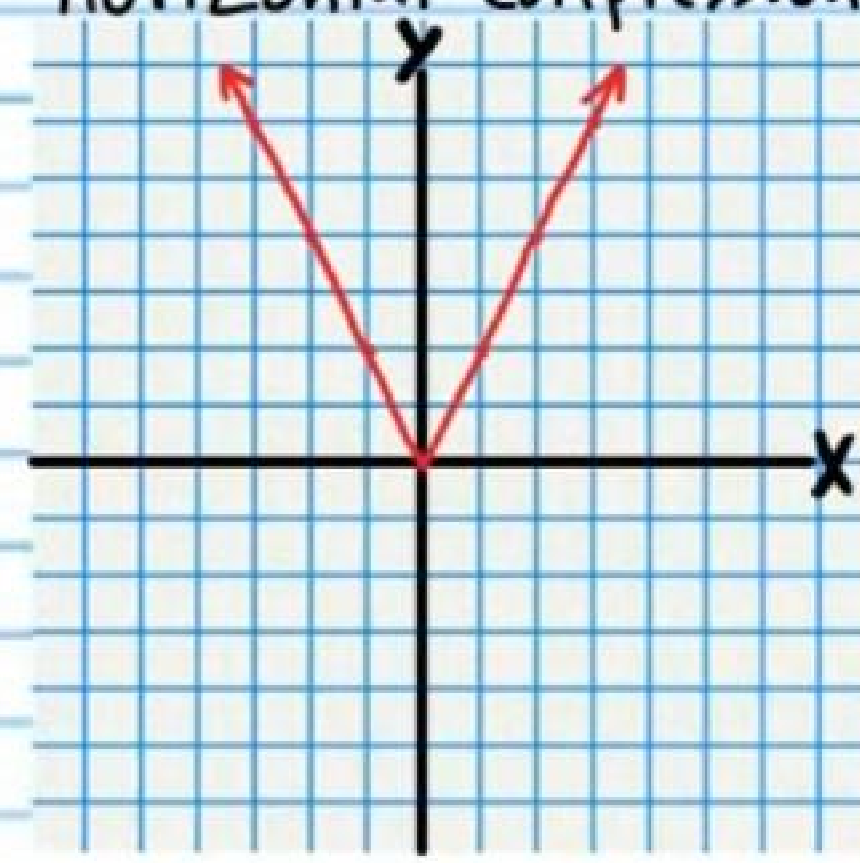
⑩ Absolute Value, $y=|x|$



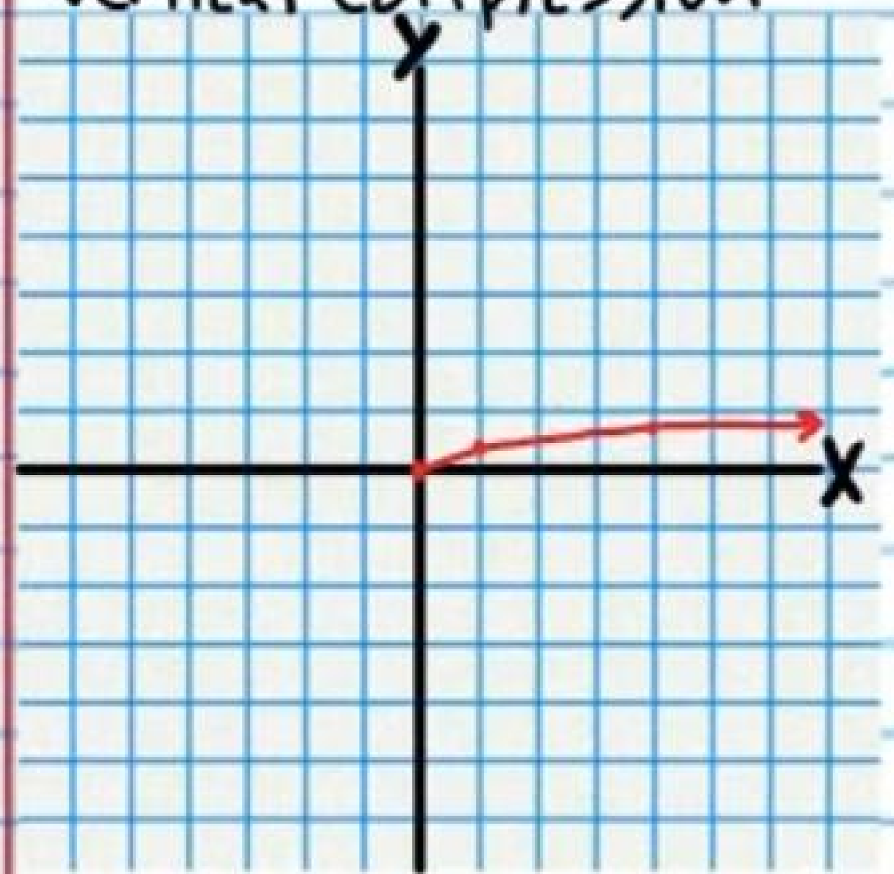
⑪ Linear, $y=x$



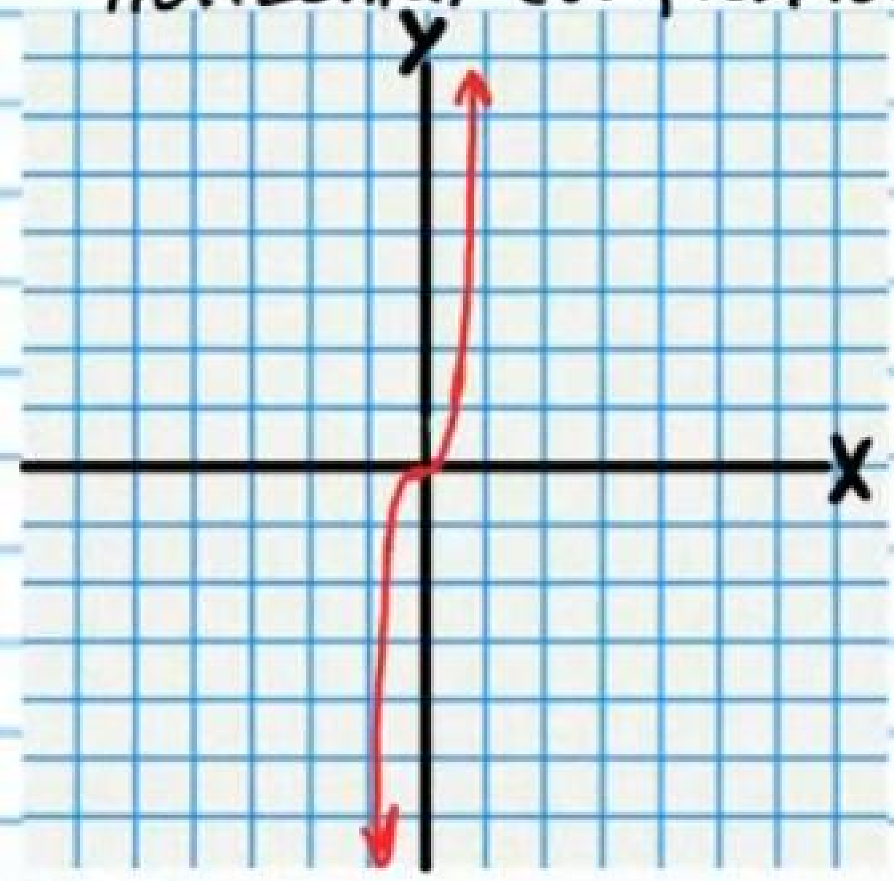
⑫ Absolute Value, $y=|x|$
Horizontal compression



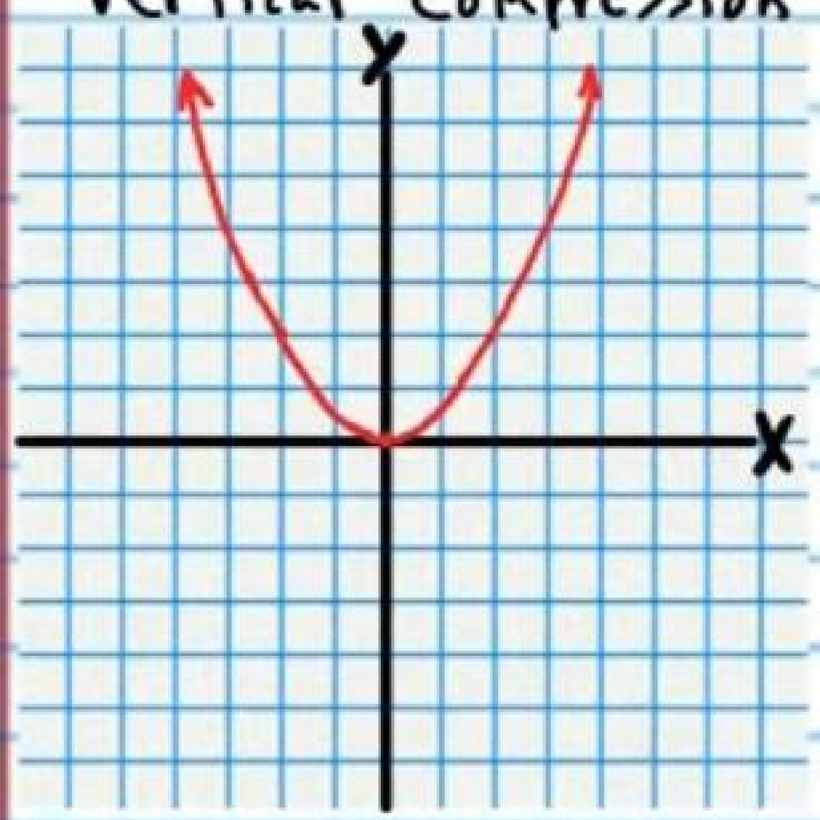
13. Square Root, $y=\sqrt{x}$
Vertical Compression



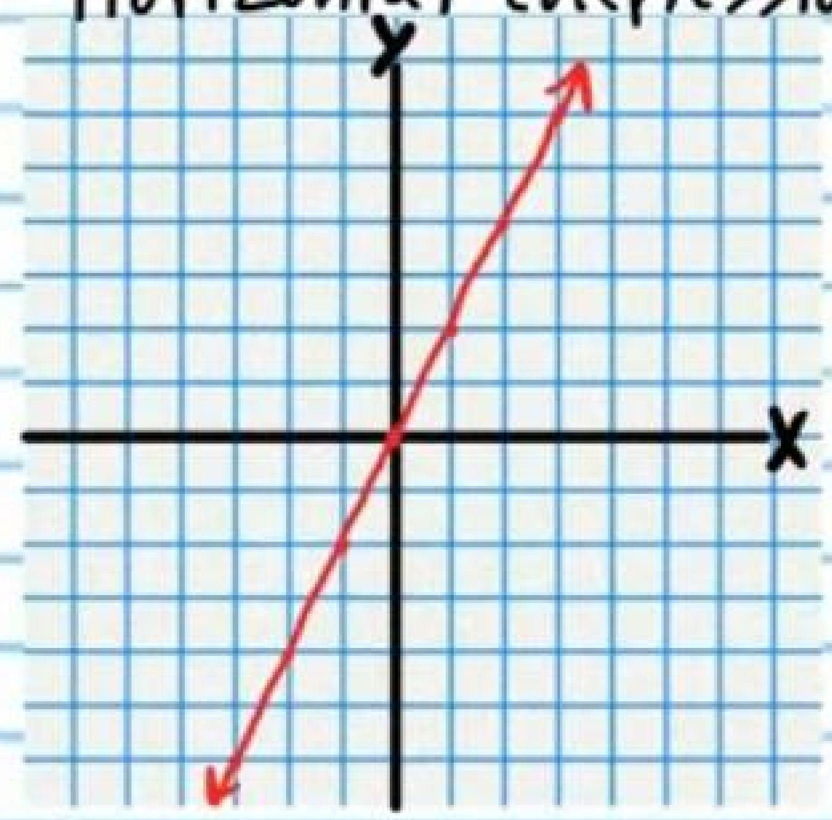
14. Cubic, $y=x^3$
Horizontal compression



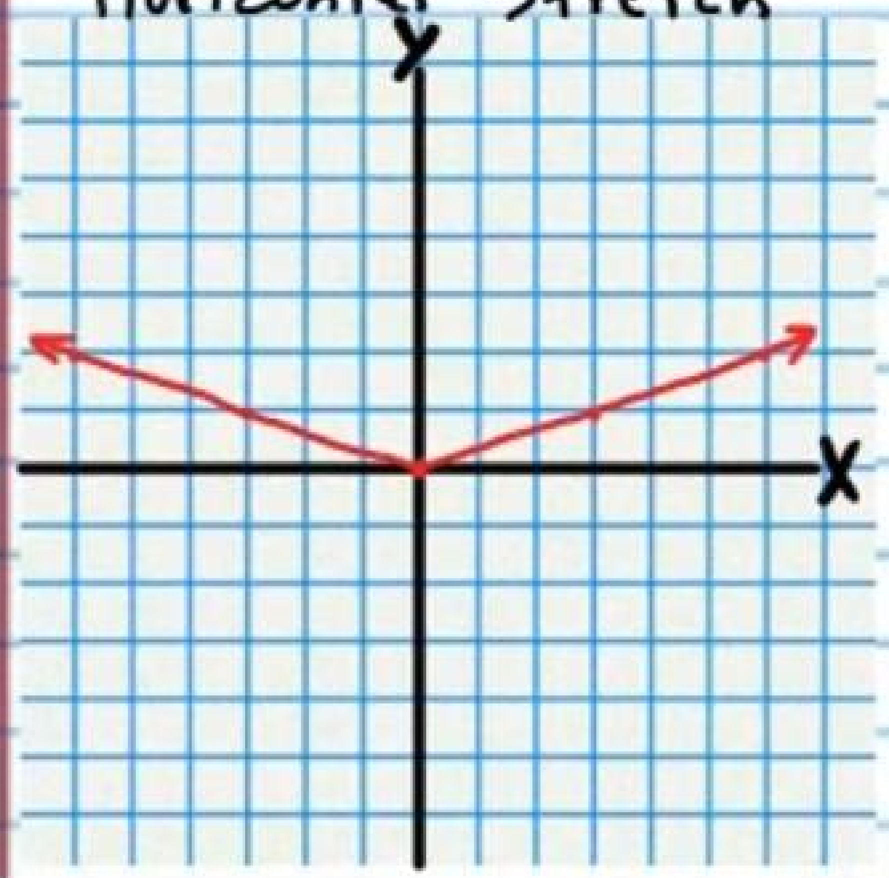
15. Quadratic, $y=x^2$
Vertical Compression



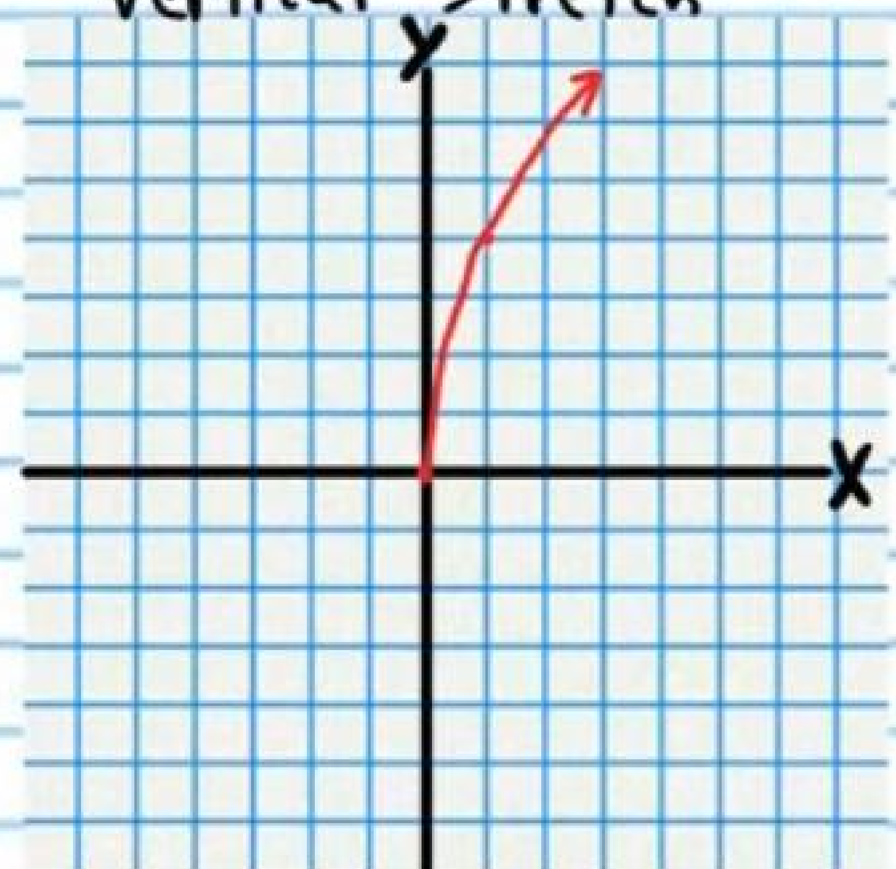
16. Linear, $y=x$
Horizontal compression



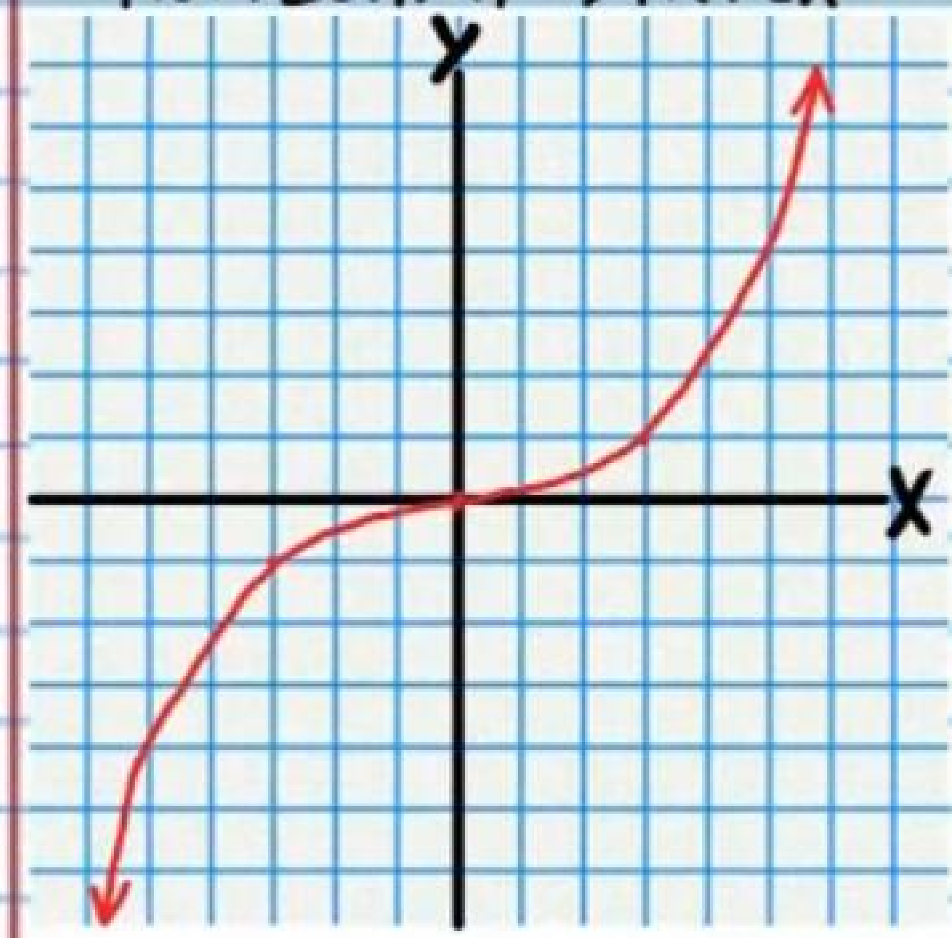
17. Absolute Value, $y=|x|$
Horizontal stretch



18. Square Root, $y=\sqrt{x}$
Vertical Stretch



19. Cubic, $y=x^3$
Horizontal stretch



20. Quadratic, $y=x^2$
Vertical Stretch